

14 (D)

A: $pV = nRT \Rightarrow T \propto pV$, $(pV)_A = 8000$, $(pV)_B = 18000$, so temp of B > A, so internal energy changes

B: gas expands, so work is done by gas

C: From A, average k.e. for B > A

D: $q = \Delta U - w = +ve - (-work\ done\ by\ gas) > 0$

15 (B)